LESSON PLAN

AIRCRAFT MAINTENANCE ORDERS (1)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	12
Total No. of Lesson Plans	16	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Loading of aircraft and determination of center of gravity, effects of variation of CG on load arm.
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Loading of aircraft and determination of center of gravity, effects of variation of CG on load arm.
5. Specific Objectives	1.1.3.1 Trainee will be able to understand modification, life categories and flight-testing effect of CG variation and wartime maintenance procedure.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Load sheet calculation exercise; Safety: Verify data accuracy; Supervision: Instructor checks arithmetic.
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What is the standard formula for determining the center of gravity?
(b) Why is the center of gravity critical for flight safety?
(c) How does the load arm relate to the moment calculation?

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Loading of aircraft procedures • Determination of center of gravity • Effects of CG variation on load arm • Modification and life categories • Flight-testing effect of CG variation	L & Ds	PPT, IB	15 min
(b)	• Wartime maintenance procedure • Authority to employ wartime procedures • Schedule maintenance requirements • Component life defect rectification • Impact of CG on maintenance limits	L & Ds	PPT, PDF	20 min
(c)	• Practical load sheet calculation • Verification of CG limits • Analysis of load arm changes • Documentation of maintenance actions • Review of wartime maintenance authority	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain how a forward shift in CG affects the load arm and aircraft stability.
(b) Describe the authority required to employ a wartime maintenance procedure for CG defects.
(c) What are the flight-testing implications of exceeding CG modification limits?

12. Home Assignment and Follow Up	Prepare a brief report on the schedule maintenance requirements for component life defect rectification related to CG variations, referencing the procedures covered in the next lesson plan (LP 13).
Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Wartime maintenance, procedure, authority to employ war time maintenance procedure schedule maintenance, component life defect rectification and repair, BDR recovery to peace time documentation.	Refer to the master syllabus reference list	LP 13

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min